

Name

key

Show work. Real solutions required. Don't compute eigenvalues or solve when an Augmented Matrix (AM) is needed.

1. Solve the ODE $y'(t) + 6t^2 y(t) = e^{-2t^3}$ with $y(0) = -5$.

$$y(t) = \frac{t-5}{e^{2t^3}}$$

"Gen Sol"

$$y(0) = -5$$

$$v(t) = e^{\int 6t^2 dt} = e^{2t^3}$$

$$e^{2t^3} y' + 6t^2 e^{2t^3} y = e^{2t^3} e^{-2t^3} = 1$$

$$(e^{2t^3} y)' = 1$$

$$e^{2t^3} y = t + C$$

$$e^0(-5) = 0 + C \quad C = -5$$

2.

2.1. Write down an AM you would row reduce for the ODE $y' = Ay + b$ with $y(0) = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$

$$y_p = \begin{pmatrix} 3 + 3 \cos(t) \\ 2 \sin(t) + e^{2t} \end{pmatrix} \text{ and } y_c = c_1 e^{3t} \begin{pmatrix} \cos(2t) \left(\frac{1}{4} \right) - \sin(2t) \left(\frac{-2}{1} \right) \\ \sin(2t) \left(\frac{1}{4} \right) + \cos(2t) \left(\frac{-2}{1} \right) \end{pmatrix}$$

$$\text{AM} = \left[\begin{array}{cc|cc} 1 & -2 & 1 & 2-3 \\ 4 & 1 & 1 & -3-1 \end{array} \right]$$

$$c_1 \begin{pmatrix} 1 \\ 4 \end{pmatrix} + c_2 \begin{pmatrix} -2 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

2.2. Compute the solution of $y' = Ay + e^{2t} \begin{pmatrix} 0 \\ 2 \end{pmatrix}$ where A is belowIn[]: MatrixForm[Eigensystem[A = $\begin{pmatrix} 2 & 2 \\ 2 & -1 \end{pmatrix}$]]

Out[]: MatrixForm =

$$\begin{pmatrix} 3 & -2 \\ (2, 1) & (-1, 2) \end{pmatrix}$$

$$y(t) = c_1 e^{3t} \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2 e^{-2t} \begin{pmatrix} -1 \\ 2 \end{pmatrix} + e^{2t} \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

$$y_c = c_1 e^{3t} \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2 e^{-2t} \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

$$y_p = a e^{2t} \quad y_p' = 2a e^{2t}$$

$$2a e^{2t} = A a e^{2t} + e^{2t} \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

$$(2I - A)a = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

$$\begin{bmatrix} 0 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

$$-2a_2 = 0$$

$$-2a_1 + 4a_2 = 2$$

$$\boxed{\begin{matrix} a_2 = 0 \\ a_1 = -1 \end{matrix}}$$

3.

$$\lambda^2 - 3\lambda - 4 = 0$$

$$(\lambda - 4)(\lambda + 1) = 0 \quad \lambda = +4 \text{ or } -1$$

3.1. Solve $2y'' - 6y' - 8y = 32t$

$$y = c_1 e^{4t} + c_2 e^{-t} - 4t + 3$$

$$y_c = c_1 e^{4t} + c_2 e^{-t}$$

$$y_p = at + b, \quad y_p' = a, \quad y_p'' = 0$$

$$2(0) - 6(a) - 8(at + b) = 32t + 0$$

$$-8A = 32$$

$$-6A - 8b = 0$$

$$A = -4$$

$$B = 3$$

3.2. Write down an AM for $y(0) = -2$ and $y'(0) = 5$.

$$\text{AM} = \left[\begin{array}{cc|c} 1 & 1 & -2-3 \\ 4 & -1 & 5+4 \end{array} \right]$$

$$y(0) = -2 \Rightarrow c_1 + c_2 - 0 + 3 = -2$$

$$y' = 4c_1 e^{4t} - c_2 e^{-t} - 4$$

$$y'(0) = 5 \Rightarrow 4c_1 - c_2 - 4 = 5$$

$$\lambda^2 + 2\lambda + 5 = 0 \Rightarrow (\lambda + 1)^2 + 4 = 0 \quad \lambda + 1 = \pm 2i$$

$$\lambda = -1 \pm 2i$$

4. Solve $y'' + 2y' + 5y = 10 \cos(t)$

$$y = y_c + y_p$$

$$y_c = e^{-t} (C_1 \cos(2t) + C_2 \sin(2t))$$

$$y_p = a \cos(t) + b \sin(t)$$

$$y_p' = -a \sin(t) + b \cos(t)$$

$$y_p'' = -a \cos(t) - b \sin(t)$$

$$\begin{aligned} & -a \cos(t) - b \sin(t) \\ & + 2(b \cos(t) - a \sin(t)) \\ & + 5(a \cos(t) + b \sin(t)) \\ \hline & = 10 \cos(t) + 0 \sin(t) \end{aligned}$$

$$\begin{cases} 4a + 2b = 10 \\ -2a + 4b = 0 \end{cases} \Rightarrow \boxed{a = 2b}$$

$$4(2b) + 2b = 10 \Rightarrow \boxed{b = 1}$$

$$\boxed{a = 2, b = 1}$$

$$\boxed{y_p = 2 \cos(t) + \sin(t)}$$

5. Solve $y''' - 3y' - 2y = 8e^{-2t}$

$$y = y_c + y_p$$

Expand $[(r+1)^2(r-2)]$

Simplify

$$-2 - 3r + r^3$$

$$r = -1, r = -1, r = 2$$

$$y_c = C_1 e^{-t} + C_2 t e^{-t} + C_3 e^{2t}$$

$$y_p = a e^{-2t}, y_p' = -2a e^{-2t}, y_p'' = 4a e^{-2t}, y_p''' = -8a e^{-2t}$$

$$(-8a) - 3(-2a) - 2(a) = 8 \Rightarrow -4a = 8 \Rightarrow \boxed{a = -2}$$

$$\boxed{y_p = -2e^{-2t}}$$